

WEEK 4 ASSIGNMENT

Name : Payal Vijay Rananaware

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks :-

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group

Home > Resource Manager | Resource groups

rg-adf-pipeline-payal Resource group

Export resource groups using Bicep or Terraform Help me generate Terraform for this resource group configuration. +1

Search Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Group by none

Overview Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials JSON View

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

	Name ↑	Type	Location
	adf-payal-pipeline	Data factory (V2)	Central India
	stpayaladf07	Storage account	Central India

Add or remove favorites by pressing Ctrl+Shift+F Showing 1 - 2 of 2. Display count: auto Give feedback

Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

payalrananaware2005@...
DEFAULT DIRECTORY (PAYALRAN...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for

Home > rg-adf-pipeline-payal > stpayaladf07

stpayaladf07 | Containers

Storage account

Search

«

+ Add container

↑ Upload

↻ Refresh

🗑 Delete

🔒 Change access level

🔄 Restore containers

⚙ Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	\$logs	13/7/2026, 5:32:57 am	Private	Available	-	...
☐	input-data	13/7/2026, 5:34:25 am	Private	Available	-	...
☐	output-data	13/7/2026, 6:34:39 am	Private	Available	-	...

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

payalrananaware2005@...
DEFAULT DIRECTORY (PAYALRAN...

Home > rg-adf-pipeline-payal > stpayaladf07 | Containers

input-data

Container

Search

«

+ Add Directory

↑ Upload

🔒 Change access level

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔄 Acquire lease

...

input-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Sample - Superstore....	13/7/2026, 7:44:38 am	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Add or remove favorites by pressing Ctrl+Shift+F

Home > rg-adf-pipeline-payal > stpayaladf07 | Containers

output-data

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🏷 Rename 🔑 Acquire lease ⋮

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

output-data

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 2 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - Superstore....	13/7/2026, 7:16:56 am	Hot (Inferred)	Block blob	2.18 MiB	Available
☐	superstore_output.csv	13/7/2026, 7:44:53 am	Hot (Inferred)	Block blob	2.18 MiB	Available

Task 3: ADF Basics

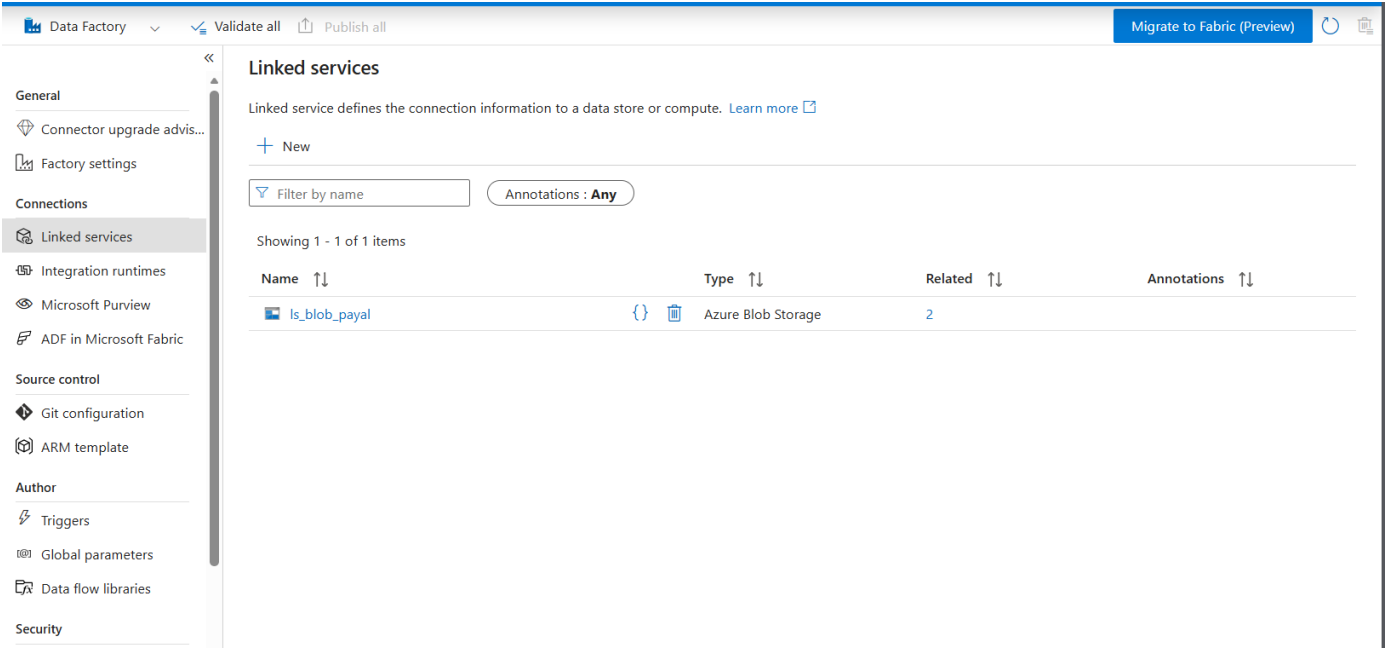
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

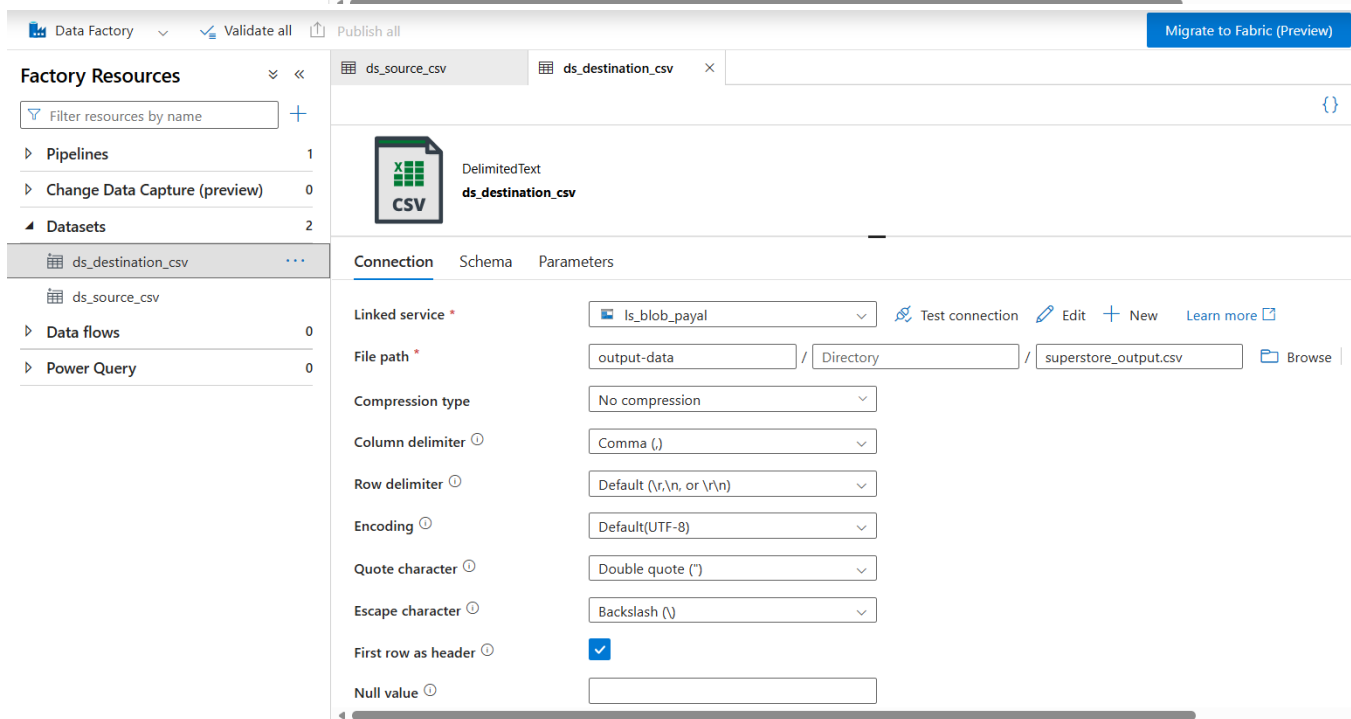
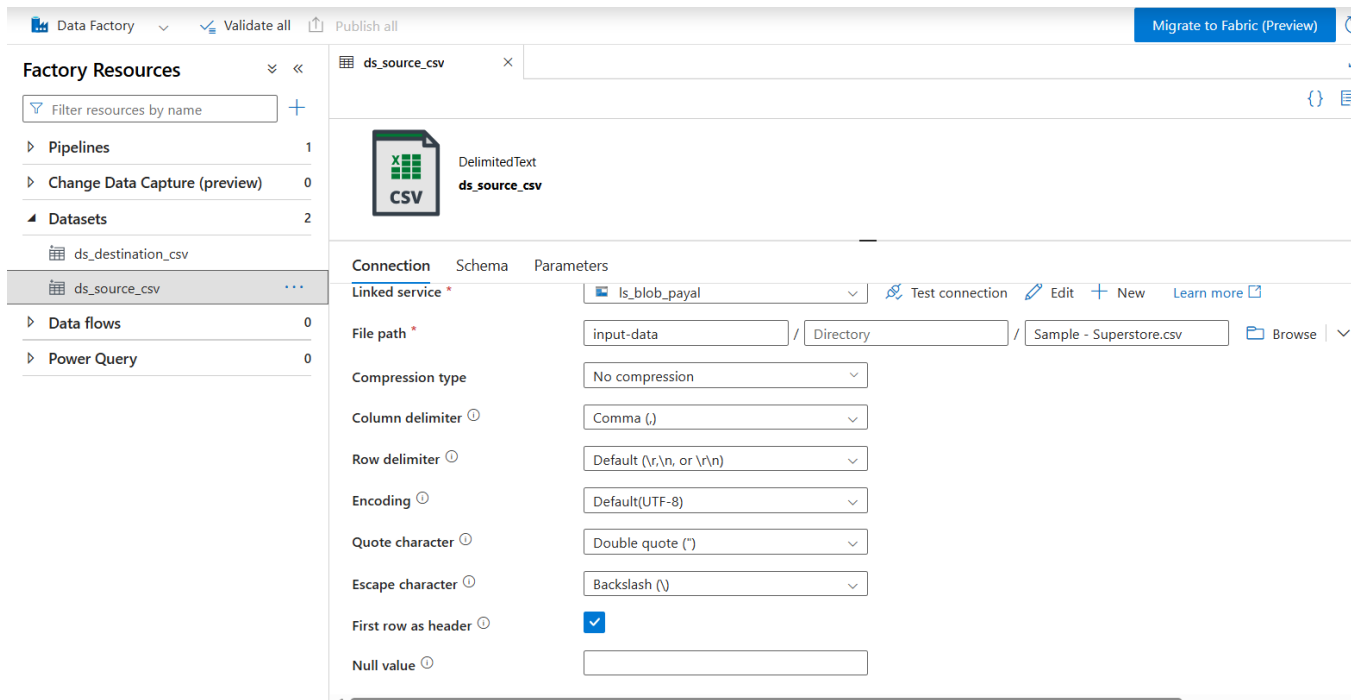
Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

Screenshot of Linked Service :-



Screenshot of dataset :-



Screenshot of Get Metadata activity :-

Data Factory | Validate all | Publish all | [Migrate to Fabric \(Preview\)](#)

Factory Resources

- Pipelines 1
 - pl_copy_data
- Change Data Capture (preview) 0
- Datasets 2
 - ds_destination_csv
 - ds_source_csv
- Data flows 0
- Power Query 0

ds_source_csv | ds_destination_csv | pl_copy_data

Validate | Debug | Add trigger

Get Metadata | Copy data

Get Metadata1 | Copy data3

General | **Settings** | User properties

Dataset * ds_source_csv [Open](#) [+ New](#) [Learn more](#)

Field list *

- ☐ Argument
- ☐ Last modified
- ☐ Size

Filter by last modified ⓘ

Start time (UTC) End time (UTC)

Skip line count

Output

[Copy to clipboard](#)

```
{
  "lastModified": "2026-07-13T03:22:36Z",
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 0
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

The ForEach activity was not used in the pipeline as the dataset contained only a single CSV file. The ForEach activity is typically used when multiple files need to be processed iteratively, which was not required in this case.

The screenshot shows the Azure Data Factory interface. The top navigation bar includes 'Data Factory', 'Validate all', 'Publish all', and 'Migrate to Fabric (Preview)'. Below the navigation bar, the pipeline 'pl_copy_data' is selected. The pipeline design canvas shows two activities: 'Get Metadata1' and 'Copy data3', connected by a dependency arrow. The 'Output' tab is active, displaying the pipeline run ID 'dcba62dc-2b40-4fc7-baf1-bc99360f5487' and the status 'Succeeded'. A table below shows the activity run details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Copy data3	✓ Succeeded	Copy data	7/13/2026, 9:07:41 AM	16s	AutoResolveIntegrationRuntime (Central India)		f48aa0d7-aa71-4c4b-a935-68f59002fe6f
Get Metadata1	✓ Succeeded	Get Metadata	7/13/2026, 9:07:35 AM	6s	AutoResolveIntegrationRuntime (Central India)		5c56b4f4-b364-46ee-98f2-4b5f583fa163

Output

 Copy to clipboard

```
{
  "lastModified": "2026-07-13T03:36:49Z",
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 1
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

» Pipeline runs

Triggered Debug

 Rerun

 Cancel options

 Refresh

 Edit columns

List

Gantt

 Filter by run ID or name

Chennai, Kolkata, Mu... : Last 24 hours

Pipeline name : All

Status : All

Runs : Latest runs

Triggered by : All

 Copy filters

 Export to CSV



 Add filter



Showing 1 - 1 items

Last refreshed over 15 minutes ago

☐	Pipeline name ↑↓	Run start ↑↓	Run end ↑↓	Duration	Triggered by	Status ↑↓	Run	Parameters	Annotations	Run
☐	pl_copy_data	7/13/2026, 8:53:32 AM	7/13/2026, 8:54:04 AM	33s	Manual trigger	 Succeeded	Original			234

Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

The screenshot shows the Azure portal interface for role assignments. The left sidebar displays the 'Storage center' and 'Resources' sections. The main pane shows the 'Role assignments' tab for the 'stpayaladf07' storage account. It indicates 4 role assignments for this subscription. The table lists the following assignments:

Name	Type	Role	Scope	Condition
Owner (2)				
Reader (1)				
adf-payal-pipeline (84ce99a5-c331-46d6-...)	Managed identity	Reader	This resource	None
Storage Blob Data Contributor (1)				
adf-payal-pipeline (84ce99a5-c331-46d6-...)	Managed identity	Storage Blob Data Contributor	This resource	Add

Showing 1 - 4 of 4 results.

Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

The screenshot displays the Azure Data Factory (ADF) console. At the top, the 'Data Factory' tab is active, showing a pipeline named 'pL_copy_data'. The pipeline is composed of two activities: 'Get Metadata1' and 'Copy data3', both of which are marked as successful with green checkmarks. Below the pipeline diagram, the 'Output' tab is selected, showing the 'Pipeline run ID' as 'dcba62dc-2b40-4fc7-baf1-bc99360f5487'. The 'Pipeline status' is 'Succeeded'. A table below the status shows the details of the two activities:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data3	Succeeded	Copy data	7/13/2026, 9:07:41 AM	16s	AutoResolveIntegrationRuntime (Central India)		f48aa0d7-aa71-4c4b-a935-68f59002fe6f
Get Metadata1	Succeeded	Get Metadata	7/13/2026, 9:07:35 AM	6s	AutoResolveIntegrationRuntime (Central India)		5c56b4f4-b364-46ee-98f2-4b5f583fa163

»

Pipeline runs

Triggered

Debug

Rerun

Cancel options

Refresh

Edit columns

List

Gantt

Filter by run ID or name

Chennai, Kolkata, Mu... : Last 24 hours

Pipeline name : All

Status : All

Runs : Latest runs

Triggered by : All

Copy filters

Export to CSV

Add filter

X

Showing 1 - 1 items

Last refreshed over 15 minutes ago

☐	Pipeline name ↑↓	Run start ↑↓	Run end ↑↓	Duration	Triggered by	Status ↑↓	Run	Parameters	Annotations	Run
☐	pl_copy_data	7/13/2026, 8:53:32 AM	7/13/2026, 8:54:04 AM	33s	Manual trigger	Succeeded	Original			234

Output

Copy to clipboard

```
{
  "lastModified": "2026-07-13T03:36:49Z",
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 1
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

Summary

- Implemented an end-to-end Azure Data Factory pipeline using Blob Storage.
- Created Resource Group, Storage Account, Blob containers, Linked Service, and datasets.
- Used Get Metadata for file validation and Copy Data for transferring CSV data from source to destination.
- Successfully executed the pipeline and verified the output file.
- Configured IAM roles to provide required access between Azure Data Factory and Storage.

Conclusion

The Azure Cloud Fundamentals and Data Pipeline Implementation task was successfully completed using Azure Storage Account and Azure Data Factory. An end-to-end data pipeline was created to transfer CSV data from Blob Storage to a destination location using Copy Data activity, with Get Metadata activity used for file validation. The pipeline was executed successfully, and the required access permissions were configured using IAM roles. Through this implementation, I gained practical understanding of Azure cloud resources, data pipeline development, monitoring, and secure data access management.